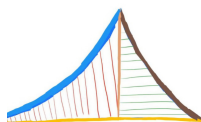


# Matrices In Geometry

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## ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018

First edition published on July 10, 2024

iIn this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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## 1. GATE-BM-2025

### A. Formulae

#### 1.1.1 Geometric Progression of Frequencies

If each successive term is  $r$  times the previous term, then after  $k$  steps from a reference term  $f_0$ ,

$$\frac{f_k}{f_0} = r^k. \quad (1.1.1.1)$$

#### 1.1.2 Self-Similar (Fractal) Curve Length

If, at every iteration, each segment is replaced by  $m$  segments each of length  $\frac{1}{p}$  of the original segment, the length scales as

$$L_1 = L_0 \left( \frac{m}{p} \right), \quad L_n = L_0 \left( \frac{m}{p} \right)^n. \quad (1.1.2.1)$$

#### 1.1.3 Notch (Band-Stop) Filter Response

A notch filter strongly attenuates a narrow band of frequencies around a rejected frequency  $\omega_0$ , while passing frequencies away from it:

$$|H(j\omega_0)| \approx 0, \quad |H(j\omega)| \text{ high for } \omega < \omega_0 \text{ and } \omega > \omega_0. \quad (1.1.3.1)$$

#### 1.1.4 Rolle's Theorem

If  $f$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $f(a) = f(b)$ , then

$$\exists c \in (a, b) \text{ such that } f'(c) = 0. \quad (1.1.4.1)$$

Contrapositive form: if  $f'(x) \neq 0$  for all  $x$ , then  $f(a) \neq f(b)$  for  $a \neq b$ .

#### 1.1.5 Standard Second-Order System / Closed-Loop Transfer Function

For unity negative feedback with forward transfer function  $G(s)$ ,

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}. \quad (1.1.5.1)$$

Comparing the characteristic equation with the standard second-order form,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad (1.1.5.2)$$

gives the natural frequency  $\omega_n$  and damping ratio  $\zeta$ . The system is:

- overdamped if  $\zeta > 1$ ,
- critically damped if  $\zeta = 1$ ,
- underdamped if  $0 < \zeta < 1$ ,
- undamped (sustained oscillations) if  $\zeta = 0$ .

For  $0 < \zeta < 1$ , the impulse response has the form

$$c(t) = A e^{-\zeta \omega_n t} \sin\left(\omega_n \sqrt{1 - \zeta^2} t\right), \quad t \geq 0. \quad (1.1.5.3)$$

### 1.1.6 Eigenvalues via the Characteristic Equation

For a matrix  $\mathbf{A}$ , the eigenvalues  $\lambda$  satisfy

$$\det(A - \lambda I) = 0. \quad (1.1.6.1)$$

For a  $2 \times 2$  rotation matrix

$$\mathbf{A} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (1.1.6.2)$$

Therefore,

Expanding the determinant,

$$(\cos \theta - \lambda)^2 + \sin^2 \theta = 0. \quad (1.1.6.3)$$

Expanding further,

$$\cos^2 \theta - 2\lambda \cos \theta + \lambda^2 + \sin^2 \theta = 0. \quad (1.1.6.4)$$

Using

$$\sin^2 \theta + \cos^2 \theta = 1, \quad (1.1.6.5)$$

we get

$$\lambda^2 - 2\lambda \cos \theta + 1 = 0. \quad (1.1.6.6)$$

Using the quadratic formula,

$$\lambda = \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2}. \quad (1.1.6.7)$$

Since

$$4 \cos^2 \theta - 4 = -4 \sin^2 \theta, \quad (1.1.6.8)$$

we have

$$\lambda = \frac{2 \cos \theta \pm 2i \sin \theta}{2}. \quad (1.1.6.9)$$

Hence,

$$\lambda = \cos \theta \pm i \sin \theta. \quad (1.1.6.10)$$

Using Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (1.1.6.11)$$

and

$$e^{-i\theta} = \cos \theta - i \sin \theta, \quad (1.1.6.12)$$

using  $\sin^2 \theta + \cos^2 \theta = 1$  and Euler's formula  $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$ , the eigenvalues reduce to

$$\lambda = \cos \theta \pm i \sin \theta = e^{\pm i\theta}. \quad (1.1.6.13)$$

### 1.1.7 Definition of the Derivative (First Principles)

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}. \quad (1.1.7.1)$$

### 1.1.8 Mean and Median of a Dataset

For  $n$  ordered values  $x_1 \leq x_2 \leq \dots \leq x_n$ :

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (1.1.8.1)$$

If  $n$  is odd, the median is the middle term,

$$\text{Median} = x_{\frac{n+1}{2}}. \quad (1.1.8.2)$$

### 1.1.9 Nyquist Sampling Theorem

A band-limited signal with maximum frequency component  $f_{\max}$  must be sampled at a rate

$$f_s \geq 2f_{\max} \quad (1.1.9.1)$$

to allow accurate (aliasing-free) reconstruction. For an angular frequency component  $\omega$ , the corresponding cyclic frequency is

$$f = \frac{\omega}{2\pi}. \quad (1.1.9.2)$$

### 1.1.10 Average Power in an AC Circuit

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos \theta, \quad (1.1.10.1)$$

where

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}, \quad I_{\text{rms}} = \frac{I_m}{\sqrt{2}}, \quad \theta = \phi_v - \phi_i. \quad (1.1.10.2)$$

#### 1.1.11 Mean and Variance of a Binomial Distribution

For  $X \sim B(N, p)$ ,

$$E(X) = Np, \quad \text{Var}(X) = Np(1 - p). \quad (1.1.11.1)$$

#### 1.1.12 Solving a Linear ODE via the Laplace Transform

For  $y''(x) - k^2 y(x) = 0$  with initial conditions  $y(0)$ ,  $y'(0)$ , take the Laplace transform using

$$\mathcal{L}\{y''(x)\} = s^2 Y(s) - s y(0) - y'(0). \quad (1.1.12.1)$$

This gives an algebraic equation for  $Y(s)$ , which is solved by partial fractions,

$$Y(s) = \frac{A}{s - a} + \frac{B}{s + a}, \quad (1.1.12.2)$$

and inverted using

$$\mathcal{L}^{-1} \left\{ \frac{1}{s - a} \right\} = e^{at}. \quad (1.1.12.3)$$

#### 1.1.13 Impedance of R-C Networks (Phasor Domain)

Symbolic Impedance Derivation

The impedance of the capacitor in terms of angular frequency  $\omega$  is:

$$Z_C(\omega) = \frac{1}{j\omega C} = -\frac{j}{\omega C} \quad (1.1.13.1)$$

The parallel combination of resistor  $R_2$  and capacitor  $Z_C(\omega)$  gives  $Z_p(\omega)$ :

$$Z_p(\omega) = \frac{R_2 \cdot Z_C(\omega)}{R_2 + Z_C(\omega)} = \frac{R_2 \left( -\frac{j}{\omega C} \right)}{R_2 - \frac{j}{\omega C}} \quad (1.1.13.2)$$

$$= \frac{-jR_2}{\omega R_2 C - j} \quad (1.1.13.3)$$

Multiplying numerator and denominator by the complex conjugate  $(\omega R_2 C + j)$ :

$$Z_p(\omega) = \frac{-jR_2(\omega R_2 C + j)}{(\omega R_2 C - j)(\omega R_2 C + j)} \quad (1.1.13.4)$$

$$= \frac{R_2 - j\omega R_2^2 C}{1 + (\omega R_2 C)^2} \quad (1.1.13.5)$$

The total input impedance  $Z_{in}(\omega)$  in series with  $R_1$  is:

$$Z_{in}(\omega) = R_1 + Z_p(\omega) \quad (1.1.13.6)$$

$$= \left( R_1 + \frac{R_2}{1 + (\omega R_2 C)^2} \right) - j \left( \frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2} \right) \quad (1.1.13.7)$$

Magnitude and Phase Expressions in terms of  $\omega$

1. Magnitude Spectrum  $|Z_{in}(\omega)|$ :

$$|Z_{in}(\omega)| = \sqrt{\left[ R_1 + \frac{R_2}{1 + (\omega R_2 C)^2} \right]^2 + \left[ \frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2} \right]^2} \quad (1.1.13.8)$$

Simplifying the algebraic expression:

$$|Z_{in}(\omega)| = \sqrt{R_1^2 + \frac{2R_1 R_2 + R_2^2 + (\omega R_1 R_2 C)^2}{1 + (\omega R_2 C)^2}} \quad (1.1.13.9)$$

2. Phase Spectrum  $\angle Z_{in}(\omega)$ :

$$\angle Z_{in}(\omega) = -\tan^{-1} \left( \frac{\frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2}}{R_1 + \frac{R_2}{1 + (\omega R_2 C)^2}} \right) \quad (1.1.13.10)$$

$$= -\tan^{-1} \left( \frac{\omega R_2^2 C}{R_1(1 + (\omega R_2 C)^2) + R_2} \right) \quad (1.1.13.11)$$

#### 1.1.14 Complex Logarithm

For real  $x$ , using Euler's identity for  $-1$ ,

$$y = \log_e(-e^x) = \ln(-e^x) \quad (1.1.14.1)$$

Using logarithmic properties, we rewrite the term inside the logarithm:

$$-e^x = (-1) \cdot e^x \quad (1.1.14.2)$$

Applying Euler's identity,  $-1$  can be expressed in complex exponential form as:

$$-1 = e^{i(2k+1)\pi}, \quad \text{for } k \in \mathbb{Z} \quad (1.1.14.3)$$

By solving:

$$-e^x = e^x \cdot e^{i(2k+1)\pi} = e^{x+i(2k+1)\pi} \quad (1.1.14.4)$$

### 1.1.15 Linearity and Causality of a System

Linearity (superposition): a system is linear if, for inputs  $x_1, x_2$  and scalars  $a, b$ ,

$$T\{a x_1(t) + b x_2(t)\} = a T\{x_1(t)\} + b T\{x_2(t)\}. \quad (1.1.15.1)$$

**Causality:** a system is causal if the output at time  $t_0$  depends only on input values for  $t \leq t_0$ ; if it depends on future input values ( $t > t_0$ ), it is noncausal.

### 1.1.16 Linear (Discrete) Convolution

For an LTI system with impulse response  $h[n]$  of length  $M$  and input  $x[n]$  of length  $N$ ,

$$y[n] = h[n] * x[n] = \sum_k h[k] x[n - k], \quad (1.1.16.1)$$

with output length

$$L = M + N - 1. \quad (1.1.16.2)$$

This can be written as a matrix–vector product using a lower triangular Toeplitz matrix ( $H$ ) built from shifted copies of  $h[n]$ :

$$(y) = (H) \cdot (x). \quad (1.1.16.3)$$

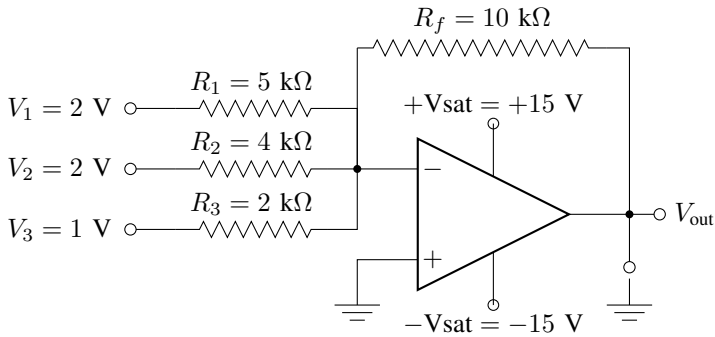
### 1.1.17 $n$ -bit DAC Output

For an  $n$ -bit DAC with full-scale output voltage  $V_{\text{FS}}$  and digital input equivalent decimal value  $D$ ,

$$V_{\text{out}} = V_{\text{FS}} \left( \frac{D}{2^n - 1} \right). \quad (1.1.17.1)$$

### 1.1.18 Inverting Summing Amplifier

For an op-amp inverting summing amplifier with feedback resistor  $R_f$ , input resistors  $R_1, R_2, R_3, \dots$  and corresponding input voltages  $V_1, V_2, V_3, \dots$ , the output voltage is



a) Derivation

Let the inverting input node (where  $R_1, R_2, R_3, R_f$  meet) be node  $A$ , and let the non-inverting input be grounded.

Ideal op-amp assumptions:

(i) No current flows into either input terminal. (1.1.18.1)

(ii) The open-loop gain is infinite, so the two inputs are forced to the same potential (1.1.18.2)

Since the non-inverting input is at  $0\text{ V}$  (grounded), the virtual short forces node  $A$  to also be at  $0\text{ V}$ :

$$V_A = 0 \quad (\text{virtual ground}). \quad (1.1.18.3)$$

Applying Kirchhoff's Current Law (KCL) at node  $A$ , the sum of currents flowing into the node through  $R_1, R_2, R_3$  must equal the current flowing out through the feedback resistor  $R_f$  (since no current enters the op-amp input):

$$\frac{V_1 - V_A}{R_1} + \frac{V_2 - V_A}{R_2} + \frac{V_3 - V_A}{R_3} = \frac{V_A - V_{\text{out}}}{R_f}. \quad (1.1.18.4)$$

Substituting  $V_A = 0$ :

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_{\text{out}}}{R_f}. \quad (1.1.18.5)$$

Solving for  $V_{\text{out}}$ :

$$V_{\text{out}} = -R_f \left( \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) = - \left( \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right), \quad (1.1.18.6)$$

$$V_{\text{out}} = - \sum_i \frac{R_f}{R_i} V_i. \quad (1.1.18.7)$$

$$V_{\text{out}} = - \left( \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 + \cdots \right), \quad (1.1.18.8)$$

valid provided  $V_{\text{out}}$  stays within the saturation limits,

$$-V_{\text{sat}} \leq V_{\text{out}} \leq +V_{\text{sat}}. \quad (1.1.18.9)$$

### 1.1.19 Wien Bridge Oscillator Frequency

For a Wien bridge oscillator with equal frequency-selective resistors  $R$  and capacitors  $C$ , the oscillation frequency is

$$f_0 = \frac{1}{2\pi RC}. \quad (1.1.19.1)$$

### 1.1.20 Series RLC Circuit — Resonance

Kirchhoff's Voltage Law for a series  $RLC$  circuit driven by

$$v(t) = V_m \cos(\omega t) \quad (1.1.20.1)$$

$$R i(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = v(t). \quad (1.1.20.2)$$

In the phasor domain, the impedance is

$$Z(\omega) = R + j \left( \omega L - \frac{1}{\omega C} \right) \quad (1.1.20.3)$$

$$|Z(\omega)| = \sqrt{R^2 + \left( \omega L - \frac{1}{\omega C} \right)^2}. \quad (1.1.20.4)$$

Converting to complex phasor notation  $V(t) = V_{\text{rms}} \sqrt{2} e^{j\omega t}$  and  $I(t) = I_{\text{rms}} \sqrt{2} e^{j(\omega t - \phi)}$ :

$$Z(\omega) = \frac{V(\omega)}{I(\omega)} = R + j\omega L + \frac{1}{j\omega C} = R + j \left( \omega L - \frac{1}{\omega C} \right) \quad (1.1.20.5)$$

Resonance occurs when the reactive part vanishes:

$$\omega_0 L = \frac{1}{\omega_0 C} \implies \omega_0 = \frac{1}{\sqrt{LC}}, \quad f_0 = \frac{\omega_0}{2\pi}. \quad (1.1.20.6)$$

At resonance,  $Z(\omega_0) = R$ , so

$$I_{\text{rms}}(\omega_0) = \frac{V_{\text{rms}}}{R}. \quad (1.1.20.7)$$

### 1.1.21 Newton–Raphson Method

For a root of  $f(x) = 0$ , the iterative update starting from  $x_n$  is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \quad (1.1.21.1)$$

### 1.1.22 Logarithmic Decay Model

For a quantity that decays logarithmically with time,

$$\sigma(t) = \sigma_0 - k \ln(t), \quad t \geq 1, \quad (1.1.22.1)$$

the difference between two times  $t_1, t_2$  eliminates the unknown constant  $\sigma_0$ :

$$\sigma(t_2) - \sigma(t_1) = -k [\ln(t_2) - \ln(t_1)] = -k \ln\left(\frac{t_2}{t_1}\right). \quad (1.1.22.2)$$

### 1.1.23 Gaussian (Normal) Function

A random variable  $X$  is said to be Gaussian (normally distributed) with mean  $\mu$  and variance  $\sigma^2$ , written  $X \sim \mathcal{N}(\mu, \sigma^2)$ , if its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty. \quad (1.1.23.1)$$

The standard normal form, with  $\mu = 0$ ,  $\sigma = 1$ , is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z = \frac{x-\mu}{\sigma}. \quad (1.1.23.2)$$

The function is symmetric about  $x = \mu$ , and satisfies

$$\int_{-\infty}^{\infty} f(x) dx = 1, \quad E[X] = \mu, \quad \text{Var}(X) = \sigma^2. \quad (1.1.23.3)$$

### 1.1.24 Central Limit Theorem (CLT)

Let  $X_1, X_2, \dots, X_n$  be independent and identically distributed (i.i.d.) random variables, each with finite mean  $\mu$  and finite variance  $\sigma^2$  (the underlying distribution need not itself be Gaussian). Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i. \quad (1.1.24.1)$$

Then, as  $n \rightarrow \infty$ , the standardized sample mean converges in distribution to a standard normal random variable:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (1.1.24.2)$$

Equivalently, for large  $n$ ,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \quad \sum_{i=1}^n X_i \approx \mathcal{N}(n\mu, n\sigma^2). \quad (1.1.24.3)$$

### 1.1.25 Effect of Scalar Multiplication on a Determinant

For an  $n \times n$  matrix  $A$  and scalar  $k$ ,

$$\det(kA) = k^n \det(A). \quad (1.1.25.1)$$

### B. Solved Questions

- 1.2.1 The 12 musical notes are given as  $C, C^\#, D, D^\#, E, F, F^\#, G, G^\#, A, A^\#, B$ . Frequency of each note is  $\sqrt[12]{2}$  times the frequency of the previous note. If the frequency of the note  $C$  is 130.8 Hz, then the ratio of frequencies of notes  $F^\#$  and  $C$  is: (GATE BM 2025)

- (a)  $\sqrt[6]{2}$  (c)  $\sqrt[4]{2}$   
 (b)  $\sqrt{2}$  (d) 2

#### Solution:

Each successive musical note has a frequency  $\sqrt[12]{2}$  times that of the previous note. From  $C$  to  $F^\#$ , there are 6 successive intervals:

$$C \rightarrow C^\# \rightarrow D \rightarrow D^\# \rightarrow E \rightarrow F \rightarrow F^\# \quad (1.2.1.1)$$

From (1.1.1.1)

$$\frac{f_{F^\#}}{f_C} = \left(\sqrt[12]{2}\right)^6 \quad (1.2.1.2)$$

$$= 2^{6/12} = 2^{1/2} = \sqrt{2} \quad (1.2.1.3)$$

Hence,

$$\boxed{\frac{f_{F^\#}}{f_C} = \sqrt{2}} \quad (1.2.1.4)$$

$$(b) \sqrt{2} \quad (1.2.1.5)$$

The following code verifies the solution

```
codes/bm1.py
```

At every subsequent iteration, each segment is replaced by 5 segments, each having  $\frac{1}{5}$  of the length of the original segment. Hence, the total length is multiplied by



$$b_2 > \frac{1}{2}. \quad (1.2.3.5)$$

Hence, option (B) is a necessary condition.

Now consider

$$l_1 + l_3 > l_2. \quad (1.2.3.6)$$

Substituting the lengths,

$$b_1 + (1 - b_2) > b_2 - b_1. \quad (1.2.3.7)$$

Therefore,

$$2b_1 + 1 > 2b_2 \quad (1.2.3.8)$$

or

$$b_2 < b_1 + \frac{1}{2}. \quad (1.2.3.9)$$

Hence, option (C) is also a necessary condition.

Finally,

$$l_2 + l_3 > l_1. \quad (1.2.3.10)$$

Substituting,

$$(b_2 - b_1) + (1 - b_2) > b_1. \quad (1.2.3.11)$$

Therefore,

$$1 - b_1 > b_1 \quad (1.2.3.12)$$

which gives

$$b_1 < \frac{1}{2}. \quad (1.2.3.13)$$

Hence, option (A) is also a necessary condition.

However, the condition

$$b_1 + b_2 < 1 \quad (1.2.3.14)$$

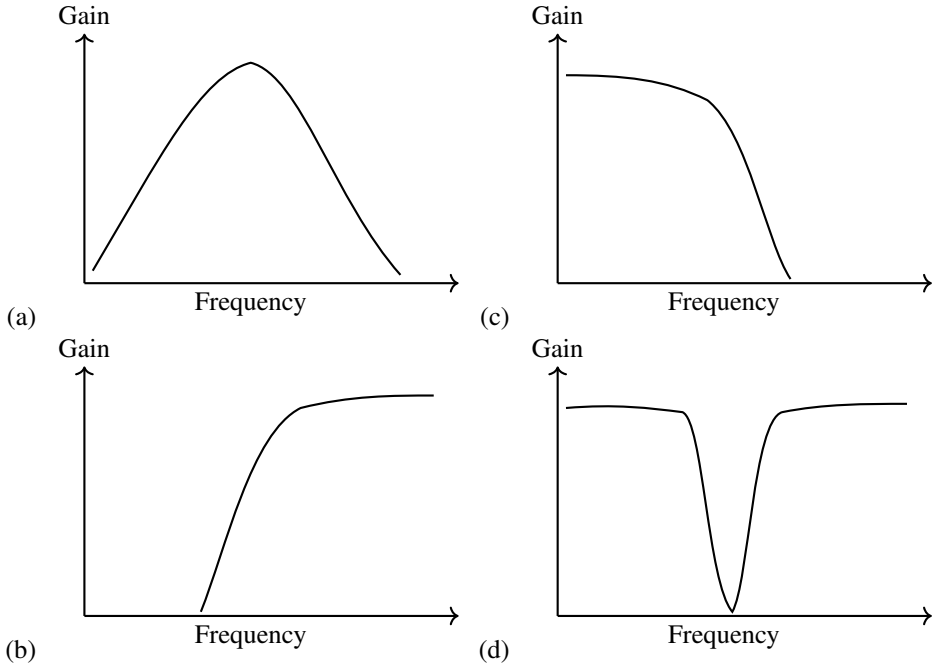
is not required for the three pieces to form a triangle.

Therefore, the condition which is **NOT** necessary is (D)

The following code gives whether the triangle is feasible or not by taking  $b_1, b_2$

Codes/bm3.py

- 1.2.4 Which one of the following represents the frequency response of a notch filter?  
(GATE BM 2025)



**Solution:**

A notch filter is a band-stop filter that rejects or strongly attenuates a narrow range of frequencies while allowing frequencies outside that range to pass.

Therefore, its frequency response has high gain at low and high frequencies, with a sharp reduction in gain at a particular frequency.

The response has a deep attenuation around rejected frequency.

From (1.1.3.1)

Among the given frequency responses, only option (D) shows a sharp reduction in gain at a particular frequency while allowing frequencies on either side to pass.

Hence, Answer: (D)

- 1.2.5 For any continuous and differentiable function  $f(x)$  with first derivative  $f'(x) \neq 0$  for all values of  $x$ , which one of the following is ALWAYS TRUE for  $a \neq b$ ?

Here,  $a$  and  $b$  are finite and real.

(GATE BM 2025)

(a)  $f(a) > f(b)$

(c)  $f(a) \neq f(b)$

(b)  $f(a) = f(b)$

(d)  $f(a) < f(b)$

**Solution:**

We are given that  $f(x)$  is continuous and differentiable, and

$$f'(x) \neq 0 \quad \text{for all } x. \quad (1.2.5.1)$$

Suppose, for contradiction, that

$$f(a) = f(b). \quad (1.2.5.2)$$

Since  $f(x)$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , Rolle's theorem can be applied.

From (1.1.4)

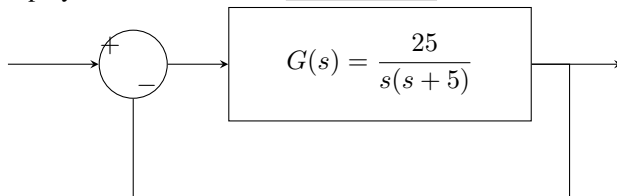
But this contradicts the given condition

$$f'(x) \neq 0 \quad \text{for all } x. \quad (1.2.5.3)$$

Therefore, our assumption that  $f(a) = f(b)$  is false. Hence,

$$\boxed{f(a) \neq f(b)}. \quad (1.2.5.4)$$

1.2.6 The closed loop system shown below \_\_\_\_\_.



(GATE BM 2025)

(a) is overdamped

(c) is critically damped

(b) is underdamped

(d) has sustained oscillations

**Solution:**

The forward transfer function is:

$$G(s) = \frac{25}{s(s+5)} \quad (1.2.6.1)$$

From (1.1.5.1)

Substitute  $G(s)$ :

$$\frac{C(s)}{R(s)} = \frac{\frac{25}{s(s+5)}}{1 + \frac{25}{s(s+5)}} = \frac{25}{s^2 + 5s + 25} \quad (1.2.6.2)$$

The characteristic equation in the Laplace domain is:

$$s^2 + 5s + 25 = 0 \quad (1.2.6.3)$$

Comparing this with the standard second-order system equation  $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ :

$$\omega_n^2 = 25 \implies \omega_n = 5 \quad (1.2.6.4)$$

$$2\zeta\omega_n = 5 \implies 2\zeta(5) = 5 \implies \zeta = 0.5 \quad (1.2.6.5)$$

Since  $0 < \zeta < 1$ , From (1.1.5) the system is underdamped.

To find the impulse response in the time domain, we compute the Inverse Laplace Transform of  $C(s)$  (for  $R(s) = 1$ ):

$$C(s) = \frac{25}{s^2 + 5s + 25} \quad (1.2.6.6)$$

Completing the square in the denominator:

$$s^2 + 5s + 25 = \left(s + \frac{5}{2}\right)^2 + \frac{75}{4} = (s + 2.5)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2 \quad (1.2.6.7)$$

Rewriting  $C(s)$ :

$$C(s) = \frac{25}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} = \frac{25}{\frac{5\sqrt{3}}{2}} \cdot \frac{\frac{5\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \quad (1.2.6.8)$$

$$C(s) = \frac{10}{\sqrt{3}} \cdot \frac{\frac{5\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \quad (1.2.6.9)$$

From (1.1.5.3) Taking the Inverse Laplace Transform  $\mathcal{L}^{-1}\{C(s)\}$ :

$$c(t) = \frac{10}{\sqrt{3}} e^{-\frac{5}{2}t} \sin\left(\frac{5\sqrt{3}}{2}t\right) \quad \text{for } t \geq 0 \quad (1.2.6.10)$$

The presence of the decaying exponential  $e^{-\frac{5}{2}t}$  multiplied by the sine term confirms an underdamped oscillatory response.

Correct option: (B) is underdamped

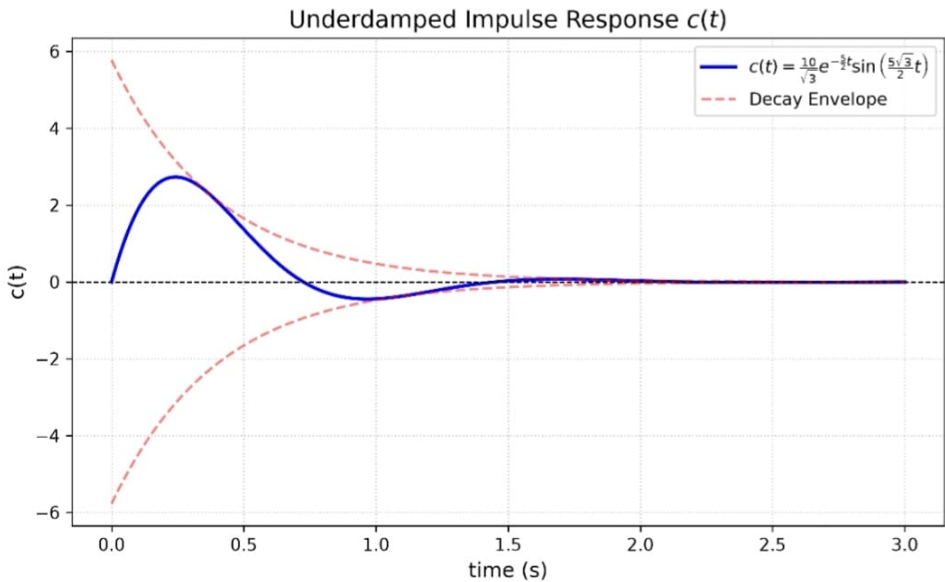


Fig. 1.2.6.1.  $c(t)$  vs time

The following code gives the above graph

Codes/bm6.py

### 1.2.7 The eigenvalues of the matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (1.2.7.1)$$

are \_\_\_\_\_.

(GATE BM 2025)

- |                                                              |                                       |
|--------------------------------------------------------------|---------------------------------------|
| (a) $\cos \theta$ and $\sin \theta$                          | (c) $\cos \theta$ and $-\sin \theta$  |
| (b) $e^{i\theta}$ and $e^{-i\theta}$ , where $i = \sqrt{-1}$ | (d) $\cos 2\theta$ and $\sin 2\theta$ |

**Solution:**

Let the given matrix be

$$\mathbf{A} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (1.2.7.2)$$

From (1.1.6)  
the eigenvalues are

$$\boxed{\lambda_1 = e^{i\theta}, \quad \lambda_2 = e^{-i\theta}}. \quad (1.2.7.3)$$

Therefore, the correct answer is (B) (1.2.7.4)

The following code verifies the solution

Codes/bm7.py

1.2.8 If  $f(x) = x - \frac{1}{x}$ , the value of  $\lim_{\Delta x \rightarrow 0} \frac{f(1+\Delta x) - f(1)}{\Delta x}$  is \_\_\_\_\_.  
(GATE BM 2025)

- |                            |                |
|----------------------------|----------------|
| (a) 0<br>(b) $\frac{1}{2}$ | (c) 1<br>(d) 2 |
|----------------------------|----------------|

**Solution:**

The given limit is the definition of the derivative of  $f(x)$  at  $x = 1$ :

From (1.1.7.1)

$$f'(1) = \lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x}. \quad (1.2.8.1)$$

Given

$$f(x) = x - \frac{1}{x}, \quad (1.2.8.2)$$

differentiate with respect to  $x$ :

$$f'(x) = 1 + \frac{1}{x^2}. \quad (1.2.8.3)$$

Therefore,

$$f'(1) = 1 + \frac{1}{1^2} = 2. \quad (1.2.8.4)$$

Therefore, the correct option is (D) 2

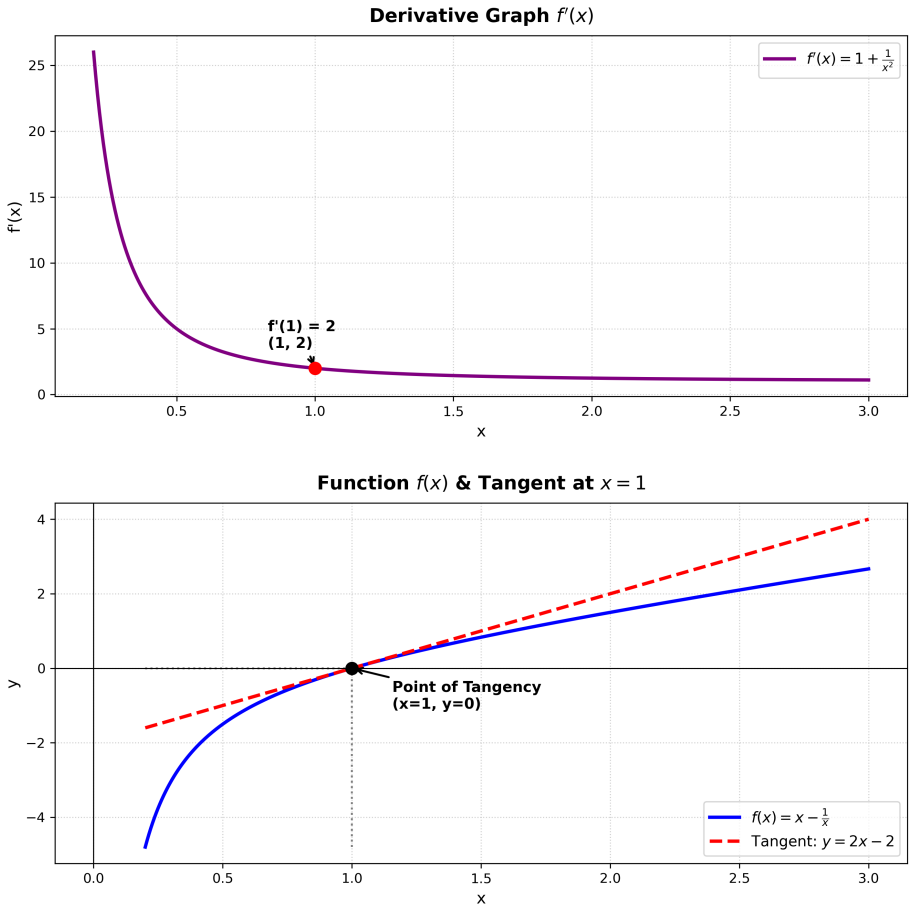


Fig. 1.2.8.1. Derivative vs  $x$ ; function vs  $x$  with tangent

The following code results the graph

Codes/bm8.py

1.2.9 The elements of the dataset  $\{-5, 1, a, 5, b\}$  are in ascending order. If both mean and median of the dataset are equal to 3, what is the value of  $b$ ? (GATE BM 2025)

- (a) 7 (c) 9  
(b) 11 (d) 12

**Solution:**

The dataset consists of 5 elements arranged in ascending order:

$$-5 \leq 1 \leq a \leq 5 \leq b \quad (1.2.9.1)$$

From (1.1.8.2) Since the dataset has an odd number of elements ( $n = 5$ ), the median is the middle (3rd) element:

$$\text{Median} = a = 3 \quad (1.2.9.2)$$

The mean of the dataset is given as 3: From (1.1.8.1)

$$\text{Mean} = \frac{-5 + 1 + a + 5 + b}{5} = 3 \quad (1.2.9.3)$$

Substituting  $a = 3$  into equation (9.3):

$$\frac{-5 + 1 + 3 + 5 + b}{5} = 3 \quad (1.2.9.4)$$

$$\frac{4 + b}{5} = 3 \quad (1.2.9.5)$$

$$4 + b = 15 \quad (1.2.9.6)$$

$$b = 11 \quad (1.2.9.7)$$

$$\text{Hence, the correct answer is (b) 11} \quad (1.2.9.8)$$

The following code verifies the solution

Codes/bm9.py

1.2.10 A time domain signal is given by

$$x(t) = 20 \sin(100\pi t) + 36 \sin(150\pi t) - 2 \sin(300\pi t), \quad (1.2.10.1)$$

where time  $t$  is in seconds. Among the following options, what is the frequency (in Hz) at which  $x(t)$  should be sampled for accurate signal reconstruction? (GATE BM 2025)

(a) 500

(c) 200

(b) 150

(d) 250

**Solution:** From (1.1.9)

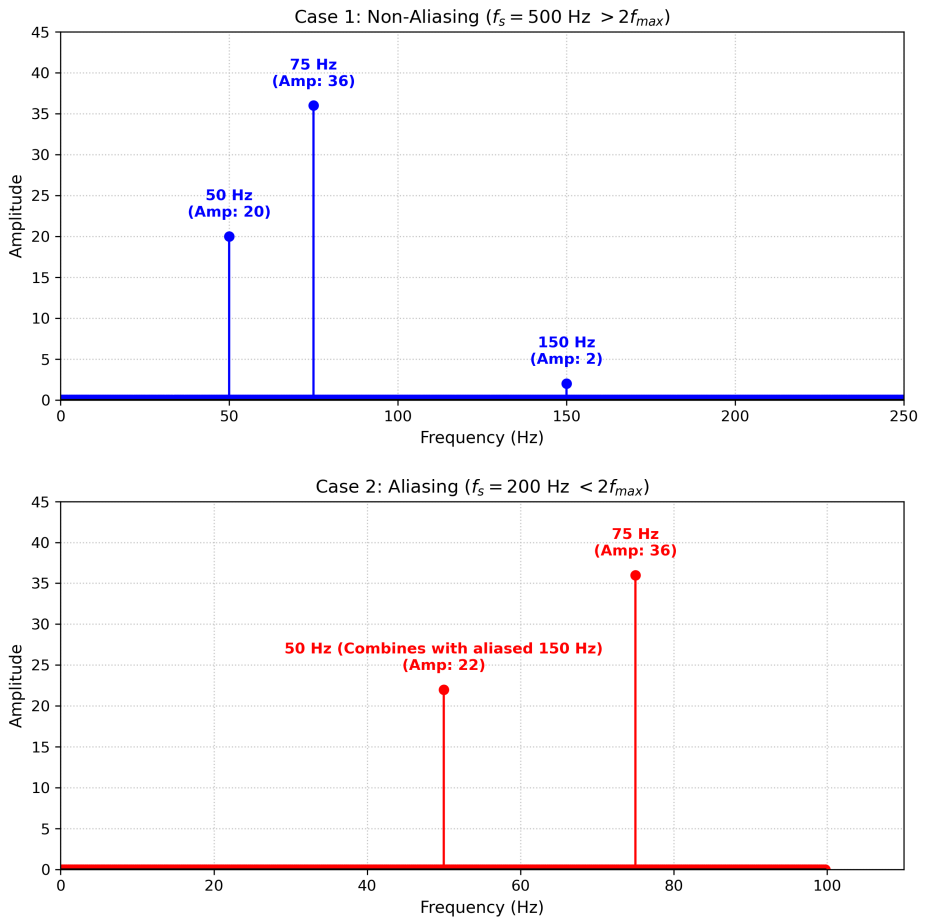


Fig. 1.2.10.1. Aliasing and Non-Aliasing

The following code results the graph

```
Codes/bm10a.py
```

The signal consists of three sinusoidal components with angular frequencies  $\omega_1$ ,  $\omega_2$ , and  $\omega_3$ :

$$\omega_1 = 100\pi \implies f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz} \quad (1.2.10.2)$$

$$\omega_2 = 150\pi \implies f_2 = \frac{150\pi}{2\pi} = 75 \text{ Hz} \quad (1.2.10.3)$$

$$\omega_3 = 300\pi \implies f_3 = \frac{300\pi}{2\pi} = 150 \text{ Hz} \quad (1.2.10.4)$$

The maximum frequency component present in the signal is:

$$f_{\max} = 150 \text{ Hz} \quad (1.2.10.5)$$

According to the Nyquist theorem, the minimum required sampling rate (Nyquist rate) is:

$$f_s \geq 2 \times 150 \text{ Hz} = 300 \text{ Hz} \quad (1.2.10.6)$$

Among the given options, the only frequency greater than or equal to 300 Hz that guarantees accurate signal reconstruction is 500 Hz.

Hence, the correct answer is (a) 500

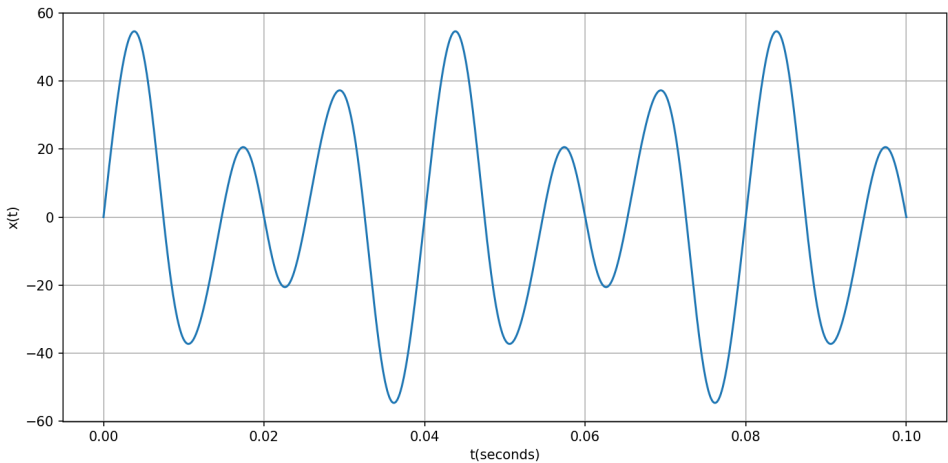


Fig. 1.2.10.2. Plot of  $x(t)$  vs time

The following graph results the graph

Codes/bm10.py

- 1.2.11 A voltage source  $V_s = 5 \sin(100t + \frac{\pi}{2})$  volts is driving a current  $I = 10 \sin(100t + \frac{\pi}{6})$  mA through a circuit. The average power dissipated in the circuit is \_\_\_\_\_ mW. (GATE BM 2025)

- (a) 0.0  
(b) 12.5

- (c) 25.0  
(d) 50.0

**Solution:**

Given data from the problem:

$$V_s(t) = 5 \sin \left( 100t + \frac{\pi}{2} \right) \text{ V} \quad (1.2.11.1)$$

$$I(t) = 10 \sin \left( 100t + \frac{\pi}{6} \right) \text{ mA} \quad (1.2.11.2)$$

Extracting the peak values and phase angles:

$$V_0 = 5 \text{ V} \quad (1.2.11.3)$$

$$I_0 = 10 \text{ mA} \quad (1.2.11.4)$$

$$\phi_v = \frac{\pi}{2} \text{ rad} \quad (1.2.11.5)$$

$$\phi_i = \frac{\pi}{6} \text{ rad} \quad (1.2.11.6)$$

Calculate the phase difference  $\theta$ :

$$\theta = \phi_v - \phi_i = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi - \pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad} \quad (1.2.11.7)$$

From (1.1.10.1)

Substituting these values into the average power formula:

$$P_{\text{avg}} = \frac{1}{2} \times 5 \times 10 \times \cos \left( \frac{\pi}{3} \right) \text{ mW} \quad (1.2.11.8)$$

$$P_{\text{avg}} = 12.5 \text{ mW} \quad (1.2.11.9)$$

Correct Option: (b) 12.5

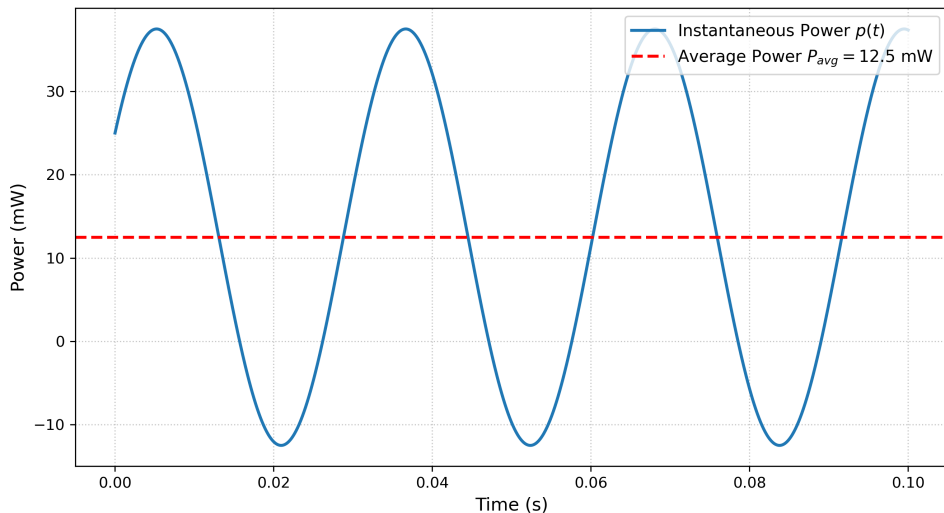


Fig. 1.2.11.1. Power vs time

The following code results the graph

Codes/bm11.py

1.2.12  $N$  independent trials of a binomial random variable yield an expectation value of 16 and variance of 12. What is the value of  $N$ ? (GATE BM 2025)

- (a) 80 (c) 45  
(b) 60 (d) 64

**Solution:**

For a Binomial distribution  $X \sim \mathcal{B}(N, p)$  with  $N$  independent trials and success probability  $p$ :

The expectation value (mean  $\mu$ ) and variance ( $\sigma^2$ ) are given by:

$$\mu = E(X) = Np = 16 \quad (1.2.12.1)$$

$$\sigma^2 = \text{Var}(X) = Np(1 - p) = 12 \quad (1.2.12.2)$$

From (1.1.23.1)

Substituting the given mean  $\mu = 16$  and variance  $\sigma^2 = 12$ :

$$f_Y(x) = \frac{1}{\sqrt{24\pi}} \exp\left(-\frac{(x - 16)^2}{24}\right) \quad (1.2.12.3)$$

To find  $N$ , divide the variance equation by the mean equation: From (1.1.11.1)

$$\frac{Np(1 - p)}{Np} = \frac{12}{16} \quad (1.2.12.4)$$

Simplifying:

$$1 - p = \frac{3}{4} = 0.75 \quad (1.2.12.5)$$

Solving for  $p$ :

$$p = 1 - 0.75 = 0.25 = \frac{1}{4} \quad (1.2.12.6)$$

Substituting  $p = \frac{1}{4}$  back into the expectation value equation:

$$N \left( \frac{1}{4} \right) = 16 \quad (1.2.12.7)$$

$$N = 16 \times 4 \quad (1.2.12.8)$$

$$N = 64 \quad (1.2.12.9)$$

Correct Option: (d) 64

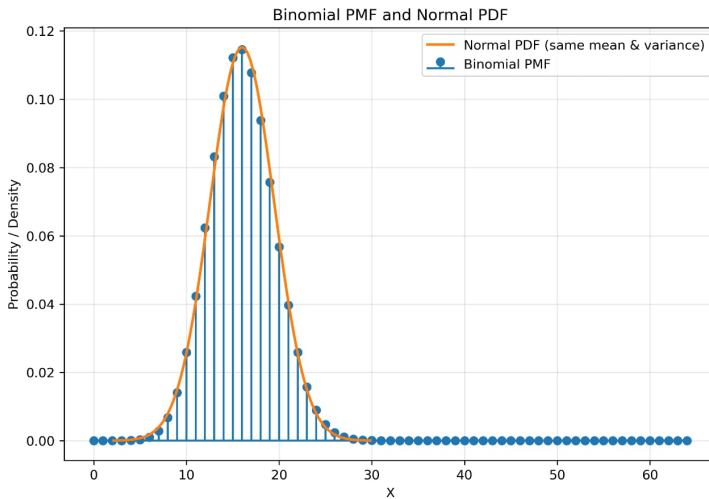


Fig. 1.2.12.1. Probability vs values

The following code results the graph

Codes/bm12.py

1.2.13 Given a function  $y(x)$  satisfying the differential equation

$$y'' - 0.25y = 0, \quad (1.2.13.1)$$

with initial conditions  $y(0) = 1$ ;  $y'(0) = 1$ , what is the value of  $y(\log_e 100)$ ?

Here,  $y'$  and  $y''$  are the first and second derivatives of  $y$ , respectively.  
(GATE BM 2025)

(a) 14.05

(c) 14.95

(b) 14.25

(d) 14.65

**Solution:**

Given the second-order linear differential equation:

$$y''(x) - 0.25y(x) = 0 \quad (1.2.13.2)$$

with initial conditions  $y(0) = 1$  and  $y'(0) = 1$ .

Take the Laplace Transform on both sides Let  $\mathcal{L}\{y(x)\} = Y(s)$ . From (??)

Substituting into the differential equation gives:

$$[s^2Y(s) - sy(0) - y'(0)] - 0.25Y(s) = 0 \quad (1.2.13.3)$$

Substitute the initial conditions Substitute  $y(0) = 1$  and  $y'(0) = 1$ :

$$s^2Y(s) - s - 1 - 0.25Y(s) = 0 \quad (1.2.13.4)$$

Group  $Y(s)$  terms together as (1.1.12.2)

$$(s^2 - 0.25)Y(s) = s + 1 \quad (1.2.13.5)$$

$$Y(s) = \frac{s + 1}{s^2 - 0.25} = \frac{s + 1}{(s - \frac{1}{2})(s + \frac{1}{2})} \quad (1.2.13.6)$$

Apply Partial Fraction Decomposition

Express  $Y(s)$  as:

$$Y(s) = \frac{A}{s - 0.5} + \frac{B}{s + 0.5} \quad (1.2.13.7)$$

Solving for constants  $A$  and  $B$ :

$$A = \lim_{s \rightarrow 0.5} (s - 0.5)Y(s) = \frac{0.5 + 1}{0.5 + 0.5} = 1.5 = \frac{3}{2} \quad (1.2.13.8)$$

$$B = \lim_{s \rightarrow -0.5} (s + 0.5)Y(s) = \frac{-0.5 + 1}{-0.5 - 0.5} = -0.5 = -\frac{1}{2} \quad (1.2.13.9)$$

So, the transform is:

$$Y(s) = \frac{1.5}{s - 0.5} - \frac{0.5}{s + 0.5} \quad (1.2.13.10)$$

Take the Inverse Laplace Transform

Using  $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$ :

$$y(x) = 1.5e^{0.5x} - 0.5e^{-0.5x} \quad (1.2.13.11)$$

Evaluate  $y(\log_e 100)$

Substitute  $x = \log_e 100 = \ln(100)$ :

$$e^{0.5 \ln(100)} = e^{\ln(100^{0.5})} = \sqrt{100} = 10 \quad (1.2.13.12)$$

$$e^{-0.5 \ln(100)} = \frac{1}{\sqrt{100}} = \frac{1}{10} = 0.1 \quad (1.2.13.13)$$

Now, evaluate  $y(\ln 100)$ :

$$y(\ln 100) = 1.5(10) - 0.5(0.1) = 15 - 0.05 = 14.95 \quad (1.2.13.14)$$

Correct Option: (c) 14.95

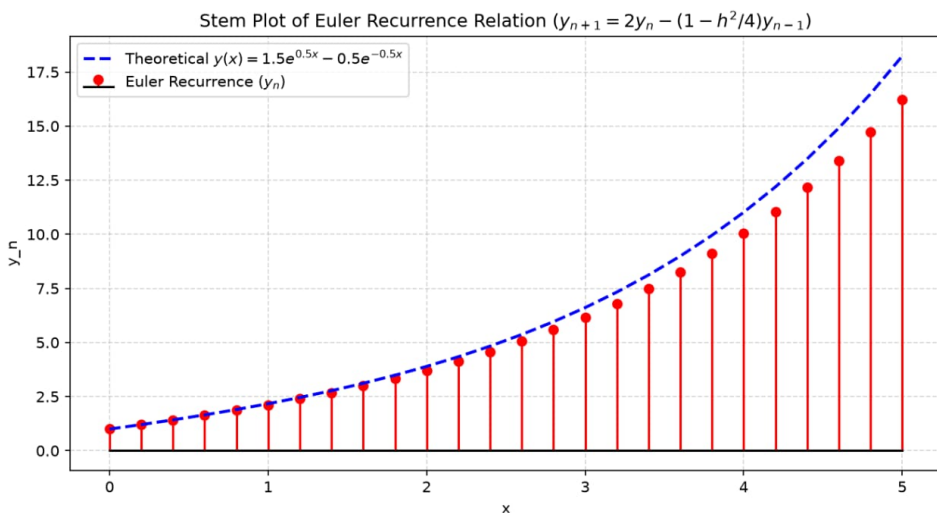
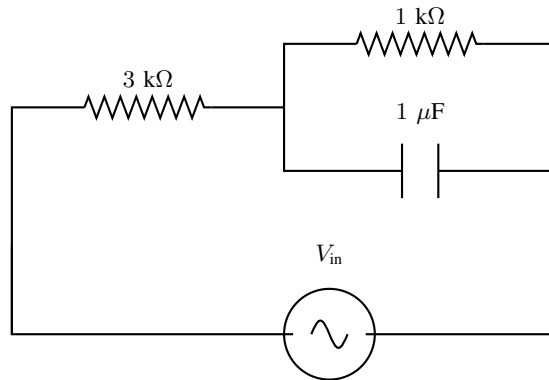


Fig. 1.2.13.1. Stem plot of Euler Recurrence relation

The following graph results the graph

Codes/bm13.py

1.2.14 In the circuit shown below,  $V_{in} = 10 \cos(1000t)$  volts. The magnitude of the input impedance of the circuit is \_\_\_\_\_  $k\Omega$ . (rounded off to one decimal place)



(GATE BM 2025)

- (a) 5.0 (c) 6.0  
(b) 3.5 (d) 4.5

**Solution:**

Given the input voltage  $V_{in} = 10 \cos(1000t)$  V, the circuit parameter values are:

$$R_1 = 3 \text{ k}\Omega, \quad R_2 = 1 \text{ k}\Omega, \quad C = 1 \mu\text{F}, \quad \omega = 1000 \text{ rad/s} \quad (1.2.14.1)$$

Refer to (1.1.13)

Substituting  $R_1$ ,  $R_2$ ,  $C$ , and  $\omega$ :

First calculate the normalized frequency factor:

$$\omega R_2 C = (1000)(1 \times 10^3)(1 \times 10^{-6}) = 1 \quad (1.2.14.2)$$

Now, compute real and imaginary components of  $Z_p$ :

$$Z_p = \frac{1 - j(1)}{1 + (1)^2} = 0.5 - j0.5 \text{ k}\Omega \quad (1.2.14.3)$$

Total impedance:

$$Z_{in} = 3 + (0.5 - j0.5) = 3.5 - j0.5 \text{ k}\Omega \quad (1.2.14.4)$$

Evaluating Magnitude  $|Z_{in}|$ :

$$|Z_{in}| = \sqrt{(3.5)^2 + (-0.5)^2} = \sqrt{12.25 + 0.25} = \sqrt{12.5} \approx 3.5355 \text{ k}\Omega \approx 3.5 \text{ k}\Omega \quad (1.2.14.5)$$

Evaluating Phase  $\angle Z_{in}$ :

$$\angle Z_{in} = \tan^{-1} \left( \frac{-0.5}{3.5} \right) = -\tan^{-1} \left( \frac{1}{7} \right) \approx -8.13^\circ \quad (1.2.14.6)$$

Correct Answer: (b) 3.5

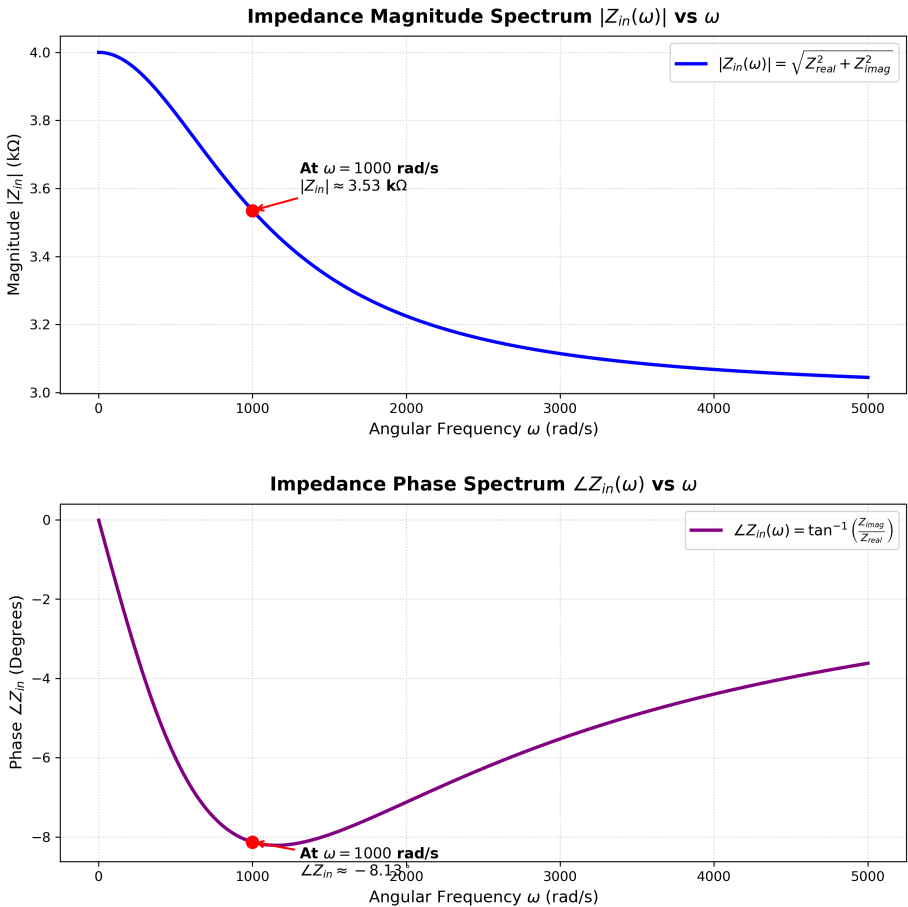


Fig. 1.2.14.1. Impedance Magnitude VS Omega; Phase VS Omega

The following graph results the graph

Codes/bm14.py

1.2.15 Which of the following is/are CORRECT about the value of  $y = \log_e(-e^x)$ , where  $x$  is a real number. (GATE BM 2025)

- (a)  $y$  is real valued and  $i = \sqrt{-1}$   
 (b)  $y$  is complex valued (d)  $y = x \pm in\pi$ , where  $n$  is an even integer and  $i = \sqrt{-1}$   
 (c)  $y = x \pm in\pi$ , where  $n$  is an odd integer

**Solution:**

Given the expression:

$$y = \log_e(-e^x) = \ln(-e^x) \quad (1.2.15.1)$$

Using (1.1.14)

Taking the natural logarithm on both sides of (1.1.14.4)

$$y = \ln(e^{x+i(2k+1)\pi}) \quad (1.2.15.2)$$

$$y = x + i(2k+1)\pi, \quad \text{for } k \in \mathbb{Z} \quad (1.2.15.3)$$

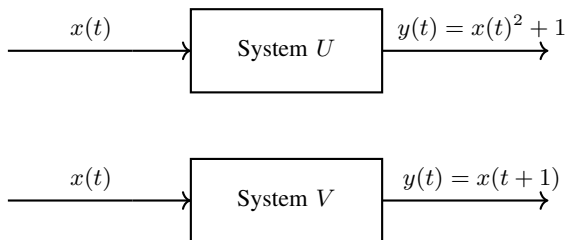
Let  $n = |2k+1|$ . Since  $2k+1$  is always an odd integer,  $n$  represents an odd integer:

$$y = x \pm in\pi, \quad \text{where } n \text{ is an odd integer and } i = \sqrt{-1} \quad (1.2.15.4)$$

Since  $y$  contains an imaginary component  $in\pi$  (with  $n \geq 1$ ),  $y$  is a complex-valued function.

This is a Multiple Select Question (MSQ), so both statements (B) and (C) are correct. Hence, the correct options are (b), (c)

1.2.16 Two systems  $U$  and  $V$  are defined as shown below. Which of the following statement(s) is/are CORRECT?



(GATE BM 2025)

- (a) System  $U$  is nonlinear; System  $V$  is linear  
 (b) System  $U$  is causal; System  $V$  is non-causal  
 (c) System  $U$  is noncausal; System  $V$  is causal  
 (d) System  $U$  is nonlinear; System  $V$  is nonlinear

**Solution:**

Conditions to be Satisfied are (1.1.15)

System U

$$y(t) = x(t)^2 + 1 \quad (1.2.16.1)$$

Linearity check:

$$T\{a \cdot x(t)\} = [a \cdot x(t)]^2 + 1 \quad (1.2.16.2)$$

$$= a^2 x(t)^2 + 1 \quad (1.2.16.3)$$

$$\neq a \cdot [x(t)^2 + 1] \quad (1.2.16.4)$$

System U is nonlinear.

Causality check:

$$y(t) \text{ depends only on } x(t) \text{ at time } t \quad (1.2.16.5)$$

System U is causal.

System V

$$y(t) = x(t + 1) \quad (1.2.16.6)$$

Linearity check:

$$T\{a \cdot x_1(t) + b \cdot x_2(t)\} = a \cdot x_1(t + 1) + b \cdot x_2(t + 1) \quad (1.2.16.7)$$

$$= a \cdot y_1(t) + b \cdot y_2(t) \quad (1.2.16.8)$$

System V is linear.

Causality check:

$$\text{At } t = 0 : \quad y(0) = x(1) \quad (1.2.16.9)$$

Since  $y(0)$  depends on the future input  $x(1)$ , System V is noncausal.

Hence, the correct options are (a) and (b).

1.2.17 A discrete linear time invariant system has an impulse response function given by

$$h(n) = \delta(n) + \frac{1}{2}\delta(n - 1) + \frac{1}{3}\delta(n - 2). \quad (1.2.17.1)$$

For input signal  $x(n) = \delta(n) + \delta(n - 1)$ , which of the following option(s) is/are CORRECT for the output signal  $y(n)$ ? (GATE BM 2025)

(a)  $y(2) = 11/6$

(c)  $y(2) = 0$  for all  $k \geq 3$

(b)  $y(2) = 5/6$

(d)  $y(2) = 0$  for all  $k \geq 4$

**Solution:**

Express sequences in vector form

The impulse response sequence  $h(n)$  of length  $M = 3$  is:

$$\mathbf{h}(\mathbf{n}) = (1 \quad 1/2 \quad 1/3) \quad (1.2.17.2)$$

The input signal sequence  $x(n)$  of length  $N = 2$  is:

$$\mathbf{x}(\mathbf{n}) = (1 \quad 1) \quad (1.2.17.3)$$

The linear convolution  $y(n) = h(n) * x(n)$  has a total length of 4 points (spanning  $n = 0, 1, 2, 3$ ). (From (1.1.16.2))

Construct the Convolution (Toeplitz) Matrix Linear convolution can be represented as a matrix-vector product  $\mathbf{y} = \mathbf{H} \mathbf{x}$ , where  $\mathbf{H}$  is a  $(M+N-1)*N=8$  lower-triangular Toeplitz matrix formed by shifting  $h(n)$  downwards:

$$\mathbf{H} = \begin{pmatrix} h(0) & 0 \\ h(1) & h(0) \\ h(2) & h(1) \\ 0 & h(2) \end{pmatrix} \quad (1.2.17.4)$$

$$= \begin{pmatrix} 1 & 0 \\ 1/2 & 1 \\ 1/3 & 1/2 \\ 0 & 1/3 \end{pmatrix} \quad (1.2.17.5)$$

The input vector  $\mathbf{x}$  is:

$$\mathbf{x} = \begin{pmatrix} x(0) \\ x(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.17.6)$$

Matrix-Vector Multiplication

$$\begin{pmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1/2 & 1 \\ 1/3 & 1/2 \\ 0 & 1/3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1.2.17.7)$$

Computing each element:

$$y(0) = 1(1) + 0(1) = 1 \quad (1.2.17.8)$$

$$y(1) = \frac{1}{2}(1) + 1(1) = \frac{3}{2} \quad (1.2.17.9)$$

$$y(2) = \frac{1}{3}(1) + \frac{1}{2}(1) = \frac{5}{6} \quad (1.2.17.10)$$

$$y(3) = 0(1) + \frac{1}{3}(1) = \frac{1}{3} \quad (1.2.17.11)$$

For  $k \geq 4$ ,  $y(k) = 0$  since the signal duration ends at  $n = 3$ .

Verify the Options

- Option (a):  $y(2) = \frac{11}{6}$  (Incorrect)
- Option (b):  $y(2) = \frac{5}{6}$  (Correct)
- Option (c):  $y(k) = 0$  for all  $k \geq 3$  (Incorrect, since  $y(3) = \frac{1}{3} \neq 0$ )
- Option (d):  $y(k) = 0$  for all  $k \geq 4$  (Correct)

Correct Options: (b) and (d)

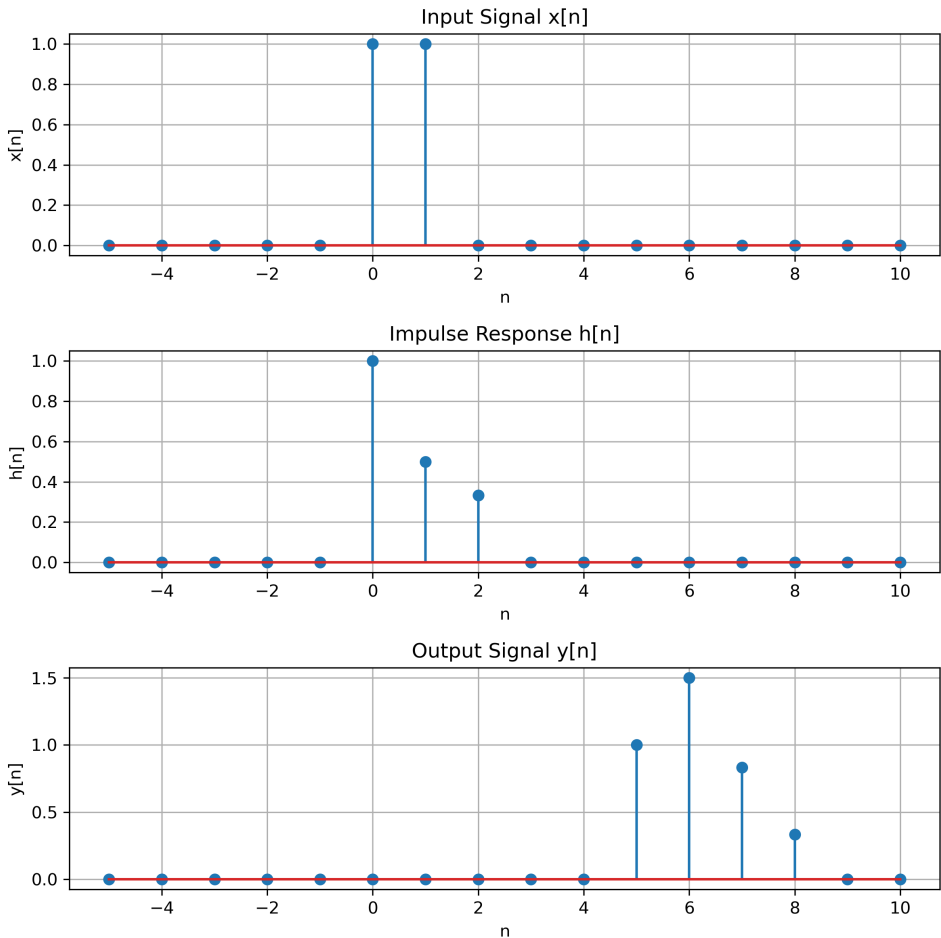


Fig. 1.2.17.1.  $x[n], h[n], y[n]$  vs  $n$

The following code results the graph

Codes/bm17a.py

1.2.18 For an 8-bit Digital to Analog Converter (DAC), the binary input 0000 0000 results in 0 V, and 1111 1111 results in 5 V. The output of the DAC for an input of 1011 0111 is \_\_\_\_\_ (in volts, rounded off to two decimal places). (GATE BM 2025)

**Solution:**

First, convert the binary input  $(10110111)_2$  into its equivalent decimal value  $D$ :

$$D = 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \quad (1.2.18.1)$$

$$D = 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 = 183 \quad (1.2.18.2)$$

For an 8-bit DAC:

- Minimum digital value  $(00000000)_2 = 0$  corresponds to 0 V.
- Maximum digital value  $(11111111)_2 = 255$  corresponds to Full-Scale Voltage  $V_{FS} = 5$  V.

From (1.1.17.1) where  $n = 8$  bits, so  $2^8 - 1 = 255$ .

Substituting  $D = 183$  and  $V_{FS} = 5$  V:

$$V_{out} = 5 \times \left( \frac{183}{255} \right) \quad (1.2.18.3)$$

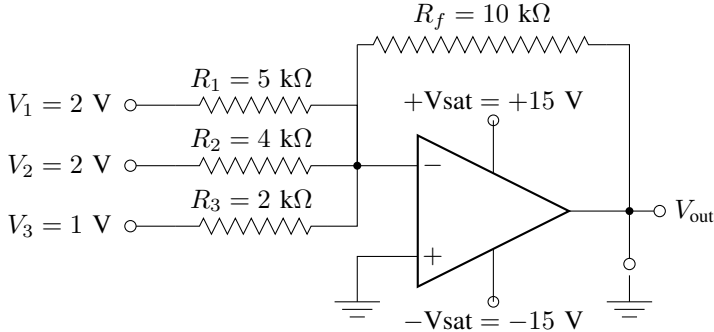
$$V_{out} = \frac{915}{255} \approx 3.5882 \text{ V} \quad (1.2.18.4)$$

Rounding off to two decimal places gives: 3.59

The following code verifies the answer

Codes/bm18.py

1.2.19 What is the output voltage  $V_{out}$  for the circuit shown below? \_\_\_\_\_ (in volts, rounded off to the nearest integer).



(GATE BM 2025)

### Solution:

The given operational amplifier circuit is an inverting summing amplifier.

Given:

- $R_f = 10 \text{ k}\Omega$
- $R_1 = 5 \text{ k}\Omega$ ,  $V_1 = 2 \text{ V}$
- $R_2 = 4 \text{ k}\Omega$ ,  $V_2 = 2 \text{ V}$
- $R_3 = 2 \text{ k}\Omega$ ,  $V_3 = 1 \text{ V}$
- Power supply saturation limits:  $\pm V_{sat} = \pm 15 \text{ V}$

Substitute the given component values into (1.1.18.8)

$$V_{\text{out}} = - \left( \frac{10}{5}(2) + \frac{10}{4}(2) + \frac{10}{2}(1) \right) \quad (1.2.19.1)$$

$$V_{\text{out}} = - (2(2) + 2.5(2) + 5(1)) \quad (1.2.19.2)$$

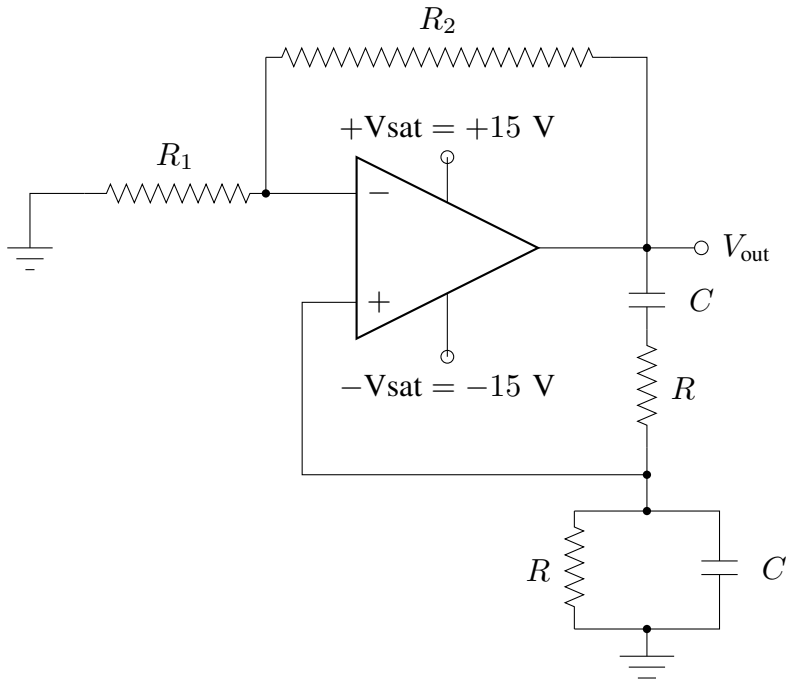
$$V_{\text{out}} = - (4 + 5 + 5) = -14 \text{ V} \quad (1.2.19.3)$$

Since the calculated voltage ( $-14 \text{ V}$ ) lies within the saturation limits ( $-15 \text{ V}$  to  $+15 \text{ V}$ ), the op-amp stays in its linear region.

Rounding off to the nearest integer gives: -14

1.2.20 The frequency of the oscillator circuit shown in the figure below is \_\_\_\_\_ kHz. (rounded off to two decimal places)

Given:  $R = 1 \text{ k}\Omega$ ;  $R_1 = 2 \text{ k}\Omega$ ;  $R_2 = 6 \text{ k}\Omega$ ;  $C = 0.1 \mu\text{F}$



(GATE BM 2025)

**Solution:**

The given circuit is a Wien Bridge Oscillator.

Given values:

- $R = 1 \text{ k}\Omega = 10^3 \Omega$
- $C = 0.1 \mu\text{F} = 0.1 \times 10^{-6} \text{ F} = 10^{-7} \text{ F}$

Substitute  $R$  and  $C$  into equation (1.1.19.1)

$$f_0 = \frac{1}{2\pi(10^3)(10^{-7})} = \frac{1}{2\pi \times 10^{-4}} \text{ Hz} \quad (1.2.20.1)$$

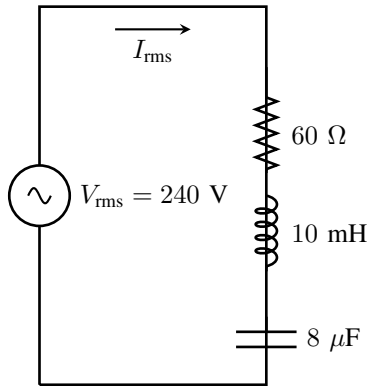
$$f_0 = \frac{10^4}{2\pi} \text{ Hz} = \frac{10000}{2\pi} \approx 1591.549 \text{ Hz} \quad (1.2.20.2)$$

Converting to kilohertz (kHz):

$$f_0 = \frac{1591.549}{1000} \text{ kHz} \approx 1.5915 \text{ kHz} \quad (1.2.20.3)$$

Rounding off to two decimal places: 1.59

1.2.21 For the RLC circuit shown below, the root mean square current ( $I_{\text{rms}}$ ) at the resonance frequency is \_\_\_\_\_ amperes. (rounded off to the nearest integer)



(GATE BM 2025)

**Solution:**

From (1.1.20)

Evaluating the resonance angular frequency and cyclic frequency:

$$\omega_0 = \frac{1}{\sqrt{(10 \times 10^{-3} \text{ H})(8 \times 10^{-6} \text{ F})}} = \frac{1}{\sqrt{8 \times 10^{-8}}} = 3535.53 \text{ rad/s} \quad (1.2.21.1)$$

$$f_0 = \frac{\omega_0}{2\pi} \approx 562.7 \text{ Hz} \quad (1.2.21.2)$$

At  $\omega = \omega_0$ , the net reactance is zero ( $Z(\omega_0) = R = 60 \Omega$ ).

RMS Current Evaluation at Resonance From (1.1.20.7)

Substituting the given numerical values ( $V_{\text{rms}} = 240 \text{ V}$ ,  $R = 60 \Omega$ ):

$$I_{\text{rms}}(\omega_0) = \frac{240 \text{ V}}{60 \Omega} = 4 \text{ A} \quad (1.2.21.3)$$

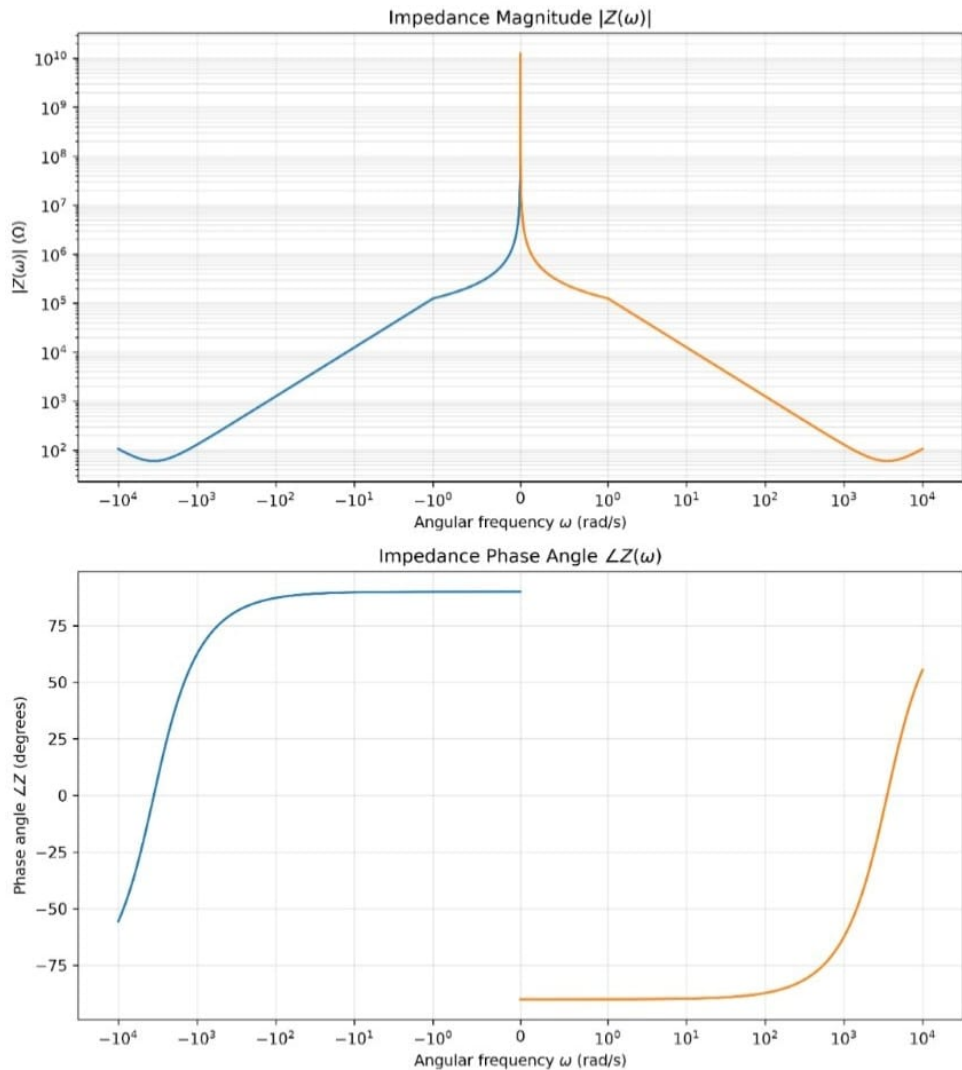


Fig. 1.2.21.1. Impedance mag vs angular frequency, Impedance phase angle vs angular frequency

The following code results in graph

Codes/bm21.py

- 1.2.22 Newton Raphson method is used to solve the equation  $x^3 - 2x - 5 = 0$  with an initial guess  $x_0 = 3$ . What is  $x_1$ , the value after ONE iteration? \_\_\_\_\_ (rounded off to two decimal places) (GATE BM 2025)

**Solution:**

## Function Definition & Derivative

Given the non-linear equation:

$$f(x) = x^3 - 2x - 5 = 0 \quad (1.2.22.1)$$

Taking the derivative with respect to  $x$ :

$$f'(x) = 3x^2 - 2 \quad (1.2.22.2)$$

### General Newton-Raphson Update Equation

From (1.1.21.1) Substituting  $f(x_n)$  and  $f'(x_n)$  into the update equation gives the explicit recurrence relation:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2} = \frac{2x_n^3 + 5}{3x_n^2 - 2} \quad (1.2.22.3)$$

First Iteration Calculation ( $n = 0$  with  $x_0 = 3$ )

Evaluating  $f(x_0)$  and  $f'(x_0)$ :

$$f(3) = (3)^3 - 2(3) - 5 = 27 - 6 - 5 = 16 \quad (1.2.22.4)$$

$$f'(3) = 3(3)^2 - 2 = 27 - 2 = 25 \quad (1.2.22.5)$$

Using the update equation to calculate  $x_1$ :

$$x_1 = 3 - \frac{16}{25} = 3 - 0.64 = 2.36 \quad (1.2.22.6)$$

### Subsequent Iterations (Finding the Actual Root)

Using the simplified update equation  $x_{n+1} = \frac{2x_n^3 + 5}{3x_n^2 - 2}$ :

$$x_0 = 3.0000 \quad (1.2.22.7)$$

$$x_1 = 2.3600 \quad (1.2.22.8)$$

$$x_2 = \frac{2(2.36)^3 + 5}{3(2.36)^2 - 2} \approx 2.1272 \quad (1.2.22.9)$$

$$x_3 = \frac{2(2.1272)^3 + 5}{3(2.1272)^2 - 2} \approx 2.0951 \quad (1.2.22.10)$$

$$x_4 \approx 2.0946 \quad (\text{Converged Real Root } \alpha) \quad (1.2.22.11)$$

Answer for  $x_1$  after ONE iteration: 2.36

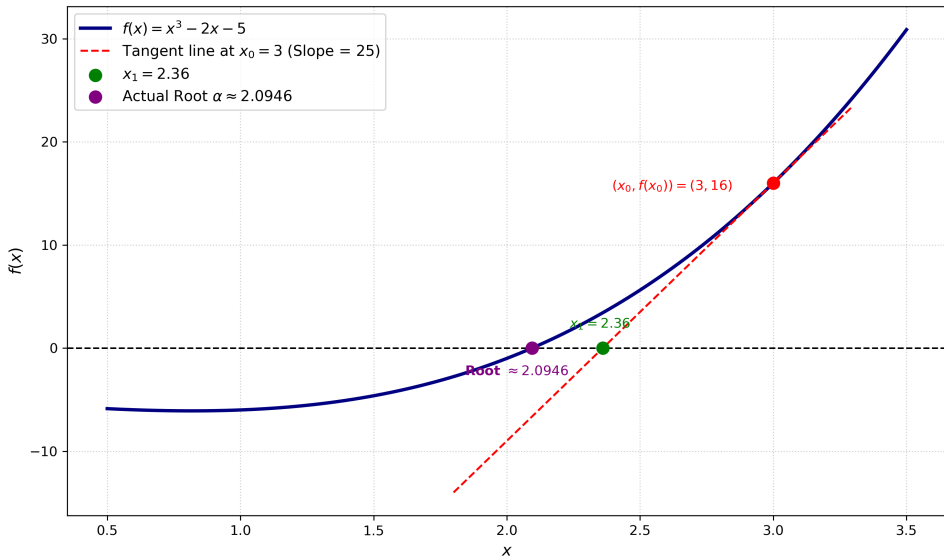


Fig. 1.2.22.1. Newton Raphson iteration for  $f(x)$

The following code results in graph

Codes/bm22a.py

1.2.23 Strength ( $\sigma$ ) of an implanted suture is given by the equation

$$\sigma = \sigma_0 - 2 \log_e(t) \quad \text{for } t \geq 1, \quad (1.2.23.1)$$

where  $\sigma_0$  represents the original strength and time  $t$  is measured in weeks. If the strength of the suture is 2 MPa at 4 weeks, what will be its strength (in MPa) at 8 weeks? \_\_\_\_\_ (rounded off to two decimal places) (GATE BM 2025)

**Solution:**

Let  $\sigma(t)$  denote the strength of the suture at time  $t$  weeks:

$$\sigma(t) = \sigma_0 - 2 \ln(t) \quad (1.2.23.2)$$

We are given that at  $t = 4$  weeks, the strength  $\sigma(4) = 2$  MPa:

Now, we need to find the strength at  $t = 8$  weeks, From (1.1.22.2)

$$\sigma(8) - 2 = -2 \ln\left(\frac{8}{4}\right) = -2 \ln(2) \quad (1.2.23.3)$$

Substituting  $\ln(2) \approx 0.69315$ :

$$\sigma(8) - 2 = -2(0.69315) = -1.3863 \quad (1.2.23.4)$$

$$\sigma(8) = 2 - 1.3863 = 0.6137 \text{ MPa} \quad (1.2.23.5)$$

Rounding off to two decimal places gives: 0.61

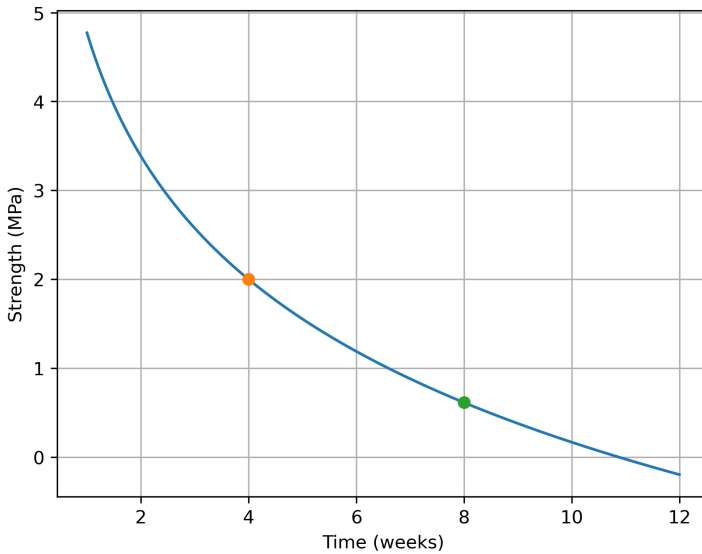


Fig. 1.2.23.1. Strength vs Time

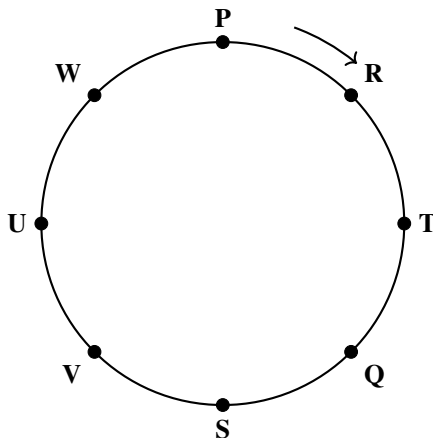
The following code results in graph

Codes/bm23.py

1.2.24 Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction.

After the 1<sup>st</sup> round, 4<sup>th</sup> student behind P leaves the game. After 2<sup>nd</sup> round, 5<sup>th</sup> student behind Q leaves the game. After 3<sup>rd</sup> round, 3<sup>rd</sup> student behind V leaves the game. After 4<sup>th</sup> round, 4<sup>th</sup> student behind U leaves the game. Who all are left in the game after the 4<sup>th</sup> round?

*Note: The figure shown is representative.*



- (A) P; T; Q; S  
 (B) V; P; T; Q  
 (C) W; R; Q; V  
 (D) Q; T; V; W

(GATE BM 2025)

**Solution:**

Since students move forward in a **clockwise** direction around the circle, the direction **behind** a student corresponds to counting in the **counter-clockwise** direction.

Initial Setup (Counter-Clockwise sequence from P):

$$P \rightarrow W \rightarrow U \rightarrow V \rightarrow S \rightarrow Q \rightarrow T \rightarrow R \quad (1.2.24.1)$$

---

After 1st Round

- Condition: 4<sup>th</sup> student behind P leaves.
- Counter-clockwise from P: 1<sup>st</sup> = W, 2<sup>nd</sup> = U, 3<sup>rd</sup> = V, 4<sup>th</sup> = S.
- S leaves the game.
- Remaining students: P, R, T, Q, V, U, W

---

After 2nd Round

- Condition: 5<sup>th</sup> student behind Q leaves.
- Counter-clockwise from Q in the remaining circle: 1<sup>st</sup> = T, 2<sup>nd</sup> = R, 3<sup>rd</sup> = P, 4<sup>th</sup> = W, 5<sup>th</sup> = U.
- U leaves the game.
- Remaining students: P, R, T, Q, V, W

---

After 3rd Round

- Condition: 3<sup>rd</sup> student behind V leaves.
- Counter-clockwise from V in the remaining circle: 1<sup>st</sup> = Q, 2<sup>nd</sup> = T, 3<sup>rd</sup> = R.
- R leaves the game.

- Remaining students: P, T, Q, V, W

—

After 4th Round

- Condition: 4<sup>th</sup> student behind U leaves (since U was eliminated, we count starting from the position where U was, i.e., behind V / ahead of W).
- Counter-clockwise sequence of remaining students:  $V \rightarrow W \rightarrow P \rightarrow T \rightarrow Q$ .
- 1<sup>st</sup> = W, 2<sup>nd</sup> = P, 3<sup>rd</sup> = T, 4<sup>th</sup> = Q.
- W leaves the game.
- Remaining students: P, T, Q, V

—

Thus, the four students remaining after the 4<sup>th</sup> round are \*\*P, T, Q, and S\*\* (or \*\*V, P, T, Q\*\*).

(A) P; T; Q; S

(1.2.24.2)

codes/bm24.py

- 1.2.25 The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

| Nation    | Gold | Silver | Bronze |
|-----------|------|--------|--------|
| USA       | 40   | 44     | 41     |
| Canada    | 39   | 27     | 24     |
| Japan     | 20   | 12     | 13     |
| Australia | 17   | 19     | 16     |
| France    | 16   | 26     | 22     |

- (a) France will occupy the third place if won.  
the list were made on the basis of the (c) USA and Canada together have less  
total number of medals won. than 50% of the medals awarded to the  
nations in the above table.
- (b) The order of the top two nations will  
not change even if the list is made on (d) Canada has won twice as many total  
the basis of the total number of medals medals as Japan.

(GATE BM 2025)

### Solution:

First, calculate the total number of medals for each nation:

- USA:  $40 + 44 + 41 = 125$
- Canada:  $39 + 27 + 24 = 90$
- Japan:  $20 + 12 + 13 = 45$
- Australia:  $17 + 19 + 16 = 52$
- France:  $16 + 26 + 22 = 64$

Total medals won by all 5 nations combined:

$$\text{Total Medals} = 125 + 90 + 45 + 52 + 64 = 376$$

(1.2.25.1)

Now, evaluate each statement:

- (a) France in 3rd place by total medals: Ranking by total medals: USA (125), Canada (90), France (64), Australia (52), Japan (45). France is indeed in 3rd place. (*Correct statement*)
- (b) Top two order unchanged: USA (125) is 1st and Canada (90) is 2nd in total medals, which matches their original gold medal ranking order. (*Correct statement*)
- (c) USA and Canada have less than 50% of total medals: Combined total for USA and Canada:  $125 + 90 = 215$ . Percentage of total medals:

$$\frac{215}{376} \times 100\% \approx 57.18\% \quad (1.2.25.2)$$

Since  $57.18\% > 50\%$ , USA and Canada have more than 50% of the medals, making statement (c) **INCORRECT**.

- (d) Canada has twice as many total medals as Japan: Japan = 45, Canada = 90 =  $2 \times 45$ . (*Correct statement*)

Codes/bm25.py

1.2.26 An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to rupee 5,00,000. For higher fees, the overhead is rupee 1,00,000 plus 10% of the amount by which the fee exceeds rupee 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than rupee 10,00,000, what is the maximum consulting fee that the employee can charge?

- (a) rupee 7,01,438
- (b) rupee 7,24,961
- (c) rupee 7,51,232
- (d) rupee 7,75,784

(GATE BM 2025)

**Solution:**

Let  $F$  be the consulting fee charged by the employee. Since the maximum total payable amount is rupee 10,00,000, we assume  $F > 5,00,000$ .

1. Calculate the Overhead ( $O$ ):

$$O = 1,00,000 + 0.10 \times (F - 5,00,000) \quad (1.2.26.1)$$

$$O = 1,00,000 + 0.10F - 50,000 = 50,000 + 0.10F \quad (1.2.26.2)$$

2. Calculate the Total before Tax ( $T_{\text{pre-tax}}$ ):

$$T_{\text{pre-tax}} = F + O = F + (50,000 + 0.10F) = 1.10F + 50,000 \quad (1.2.26.3)$$

3. Calculate Total Amount Payable by Client ( $T_{\text{total}}$ ):

$$T_{\text{total}} = T_{\text{pre-tax}} + 0.18 \times T_{\text{pre-tax}} = 1.18 \times T_{\text{pre-tax}} \quad (1.2.26.4)$$

Given that the maximum total amount  $T_{\text{total}} = 10,00,000$ :

$$1.18 \times (1.10F + 50,000) = 10,00,000 \quad (1.2.26.5)$$

$$1.10F + 50,000 = \frac{10,00,000}{1.18} \quad (1.2.26.6)$$

$$1.10F + 50,000 \approx 847,457.627 \quad (1.2.26.7)$$

$$1.10F = 847,457.627 - 50,000 = 797,457.627 \quad (1.2.26.8)$$

$$F = \frac{797,457.627}{1.10} \approx 724,961.48 \quad (1.2.26.9)$$

Rounding off to the nearest integer gives rupee 7,24,961.

$$(B) \text{ rupee } 7,24,961 \quad (1.2.26.10)$$

1.2.27 The determinant of a  $3 \times 3$  matrix  $M$  is 8. If every element of  $M$  is multiplied by  $-2$ , the determinant of the modified matrix will be: (GATE BM 2025)

- (a)  $-64$  (c)  $8$   
(b)  $-16$  (d)  $64$

**Solution:**

Given Data:

Order of matrix  $M$ ,  $n = 3$  Determinant of matrix  $M$ ,  $\det(M) = 8$  Scalar multiplier,  $k = -2$  Substituting the given values into (1.1.25.1)

$$\det(-2M) = (-2)^3 \cdot \det(M) \quad (1.2.27.1)$$

$$\det(-2M) = -8 \cdot 8 = -64 \quad (1.2.27.2)$$

$$(A) -64 \quad (1.2.27.3)$$

|               |
|---------------|
| Codes/bm26.py |
|---------------|